CS3230 Design and Analysis of Algorithms	October 26, 2019

Applications of Flows and Cuts, NP-Completeness

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Problem 9-1 (Passes Through)

Given an undirected graph $G = (V, E)$, with three vertices u, v, and w, describe and analyze an algorithm to determine whether there is a path from u to w that passes through v. (**Hint**: If G were a directed graph, this problem would be NP-hard!)

Problem 9-2 (Cycle Cover)

A cycle cover of a given directed graph $G = (V, E)$ is a set of vertex-disjoint cycles that cover every vertex in G. Describe and analyze an efficient algorithm to find a cycle cover for a given graph, or correctly report that no cycle cover exists. (**Hint**: Use bipartite matching!)

Problem 9-3 (Hardness of MAX-2SAT and MAX-CUT)

(a) We want to analyze a reduction from 3SAT to the following language:

$$\mathsf{MAX\text{-}2SAT} = \{(\phi', k) \mid \phi' \text{ is a 2-CNF formula, and there is an assignment that satisfies at least } k \text{ clauses}\}.$$

Our reduction is the following: Given a 3SAT instance ϕ, we will output a MAX-2SAT instance (ϕ', k), where ϕ' is a 2-CNF formula. To construct ϕ', do the following: for each clause $(x \vee y \vee z)$ in ϕ, add the following 10 clauses to ϕ' (where w is a fresh variable for each clause):

$$(x), (y), (z), (\neg x \vee \neg y), (\neg y \vee \neg z), (\neg x \vee \neg z), (w), (x \vee \neg w), (y \vee \neg w), (z \vee \neg w)$$

Find a value of k such that $(\phi', k) \in \mathsf{MAX\text{-}2SAT}$ if and only if $\phi \in \mathsf{3SAT}$. Prove the correctness of the reduction.

Now consider the following language:

$$\mathsf{MAX\text{-}CUT} = \{(G, c) \mid G \text{ is a multigraph with a cut of size at least } c\}.$$

A *cut* in a graph is a partition of its vertices into two parts. The size of the cut is the number of edges which "cross" the cut (whose endpoints are in opposite parts). A multigraph means we allow duplicate edges.

We now analyze a reduction from MAX-2SAT to MAX-CUT. Given an instance (ϕ, k) of MAX-2SAT, let n be the number of variables occuring in ϕ, and m the number of clauses. Consider the following graph: G_ϕ is a graph with a vertices labeled x_i and $\neg x_i$ for each

variable x occuring in ϕ, and two special vertices labeled T and F. We add $5m$ edges between T and F, and $5m$ edges between each pair $(x_i, \neg x_i)$ — see Figure 1. Then, for each clause $(x \vee y) \in \phi$, where x and y are literals, we add the following 7 edges (see clause subgraph in Figure 2): (x, y), (T, x), (T, y), and two copies of the edges (x, F) and (y, F).

(b) Show that in the largest cut in G_ϕ, T and F must be in opposite parts.

(c) Show that in the largest cut in G_ϕ, the vertices corresponding to x and $\neg x$ must be in opposite parts.

(d) Argue that $(\phi, k) \in$ MAX-2SAT if and only if $(G_\phi,\ 5m + 5mn + 4k + 2(m - k)) \in$ MAX-CUT.

Problem 9-4 (A polynomial time algorithm for 2SAT)

We want to show an efficient algorithm for 2SAT, which might be a little surprising since 3SAT is NP-hard. Given an instance ϕ of 2SAT, we first add a new literal z to each one-literal clause. We then construct a directed graph $G = (V, E)$ with vertices V labeled by the literals x and $\overline{x}$ for each variable x occuring in ϕ. This graph has an edge $(u, v) \in E$ directed from vertex u to vertex v if the clause $(\overline{u} \vee v)$ is in ϕ. If $(\overline{u} \vee v)$ is in ϕ, so is $(v \vee \overline{u})$ because of commutativity of $\vee$. Thus, if $(u, v) \in E$, then $(\overline{v}, \overline{u}) \in E$ as well (See Figure 3). Note that $(u, v) \neq (\overline{v}, \overline{u})$ because this requires that $v = \overline{u}$, which is not allowed. Let $u \neq \overline{w}$. It follows that if there is a path from u to w in G, there is a distinct path from $\overline{w}$ to $\overline{u}$) obtained by reversing the directions of each edge on the path and replacing the literals by their complements.

To understand why these edges are chosen, note that if all clauses of ϕ are satisfied and $(\overline{u} \vee v)$ is in ϕ, then $u = 1$ implies that $v = 1$. This implication relation, denoted $u \Rightarrow v$, is transitive. If there is a path $(u_1, u_2, \ldots, u_k)$ in G, then there are clauses $(\overline{u}_1 \vee u_2), (\overline{u}_2 \vee u_3), \ldots, (\overline{u}_{k-1} \vee u_k)$ in ϕ. If all clauses are satisfied and if the literal $u_1 = 1$, then each unnegated literal on this path must have value 1.

(a) Show that if there exists a variable x such that there is a path in G from x to $\overline{x}$ and from $\overline{x}$ to x, then ϕ is not satisfiable.

(b) Hence show that if there if ϕ is not satisfiable, then there must be a literal x such that there is a path from x to $\overline{x}$, and from $\overline{x}$ to x. Conclude that 2SAT can be solved in polynomial time.

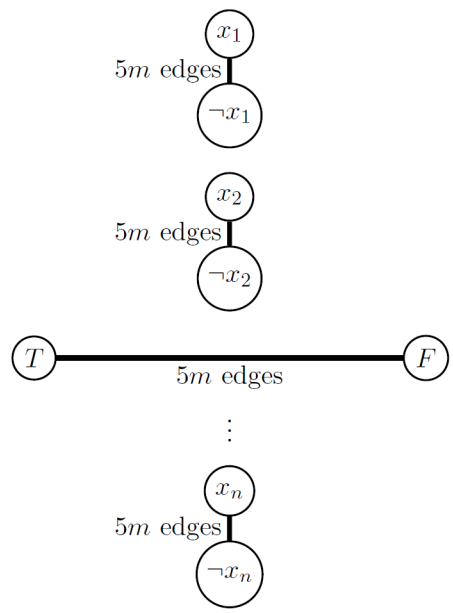

Figure 1: Starting graph for G_ϕ, where ϕ has n variables, $x_1, \ldots, x_n$.

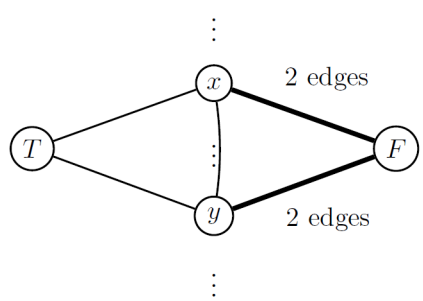

Figure 2: Edges to add for a clause of the form $(x \vee y)$.

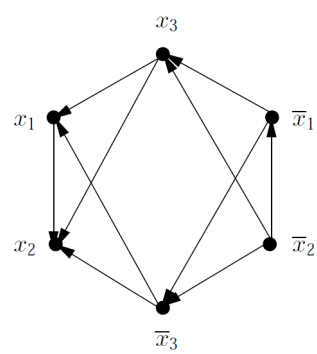

Figure 3: A graph capturing the implications associated with the following satisfiable instance of 2-SAT: $(x_3 \vee x_2) \wedge (x_3 \vee x_1) \wedge (\overline{x}_3 \vee x_2) \wedge (\overline{x}_1 \vee x_2) \wedge (\overline{x}_3 \vee x_1)$.